RAJALAKSHMI ENGINEERING COLLEGE

Hariharaa.S 251001069-2025-IT

V2

| | |
|---|---|
| **Started on** | Friday, 21 August 2026, 3:25 PM |
| **State** | Finished |
| **Completed on** | Friday, 21 August 2026, 3:45 PM |
| **Time taken** | 20 mins 58 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int m;
    if (scanf("%d", &m) != 1) return 0;

    int arr[m];
    for (int i = 0; i < m; i++) {
        if (scanf("%d", &arr[i]) != 1) return 0;
    }

    int low = 0, high = m - 1;
    int firstZeroIndex = -1;

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == 0) {
            firstZeroIndex = mid;
            high = mid - 1;
        } else {
            low = mid + 1;
        }
    }

    if (firstZeroIndex == -1) {
        printf("0\n");
    } else {
        printf("%d\n", m - firstZeroIndex);
    }

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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V2

| | |
|---|---|
| **Started on** | Friday, 21 August 2026, 4:02 PM |
| **State** | Finished |
| **Completed on** | Friday, 21 August 2026, 4:15 PM |
| **Time taken** | 13 mins 12 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

**Example 1:**

```
Input: nums = [3,2,3]
Output: 3
```

**Example 2:**

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

**For example:**

| Input | Result |
|---|---|
| 3<br>3 2 3 | 3 |
| 7<br>2 2 1 1 1 2 2 | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int majorityElement(int* nums, int numsSize) {
    int candidate = nums[0];
    int count = 0;

    for (int i = 0; i < numsSize; i++) {
        if (count == 0) {
            candidate = nums[i];
        }

        if (nums[i] == candidate) {
            count++;
        } else {
            count--;
        }
    }

    return candidate;
}

int main() {
    int n;
    while (scanf("%d", &n) != EOF) {
        int nums[n];
        for (int i = 0; i < n; i++) {
            scanf("%d", &nums[i]);
        }
        printf("%d\n", majorityElement(nums, n));
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>3 2 3 | 3 | 3 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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V2

| | |
|---|---|
| **Started on** | Friday, 21 August 2026, 4:21 PM |
| **State** | Finished |
| **Completed on** | Friday, 21 August 2026, 4:27 PM |
| **Time taken** | 6 mins 5 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

**Output Format**

First Line Contains Integer – Floor value for x

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, x;
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    scanf("%d", &x);

    int floor = -1;
    for (int i = 0; i < n; i++) {
        if (arr[i] <= x) {
            floor = arr[i];
        }
    }

    printf("%d\n", floor);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6<br>1<br>2<br>8<br>10<br>12<br>19<br>5 | 2 | 2 | ✔ |
| ✔ | 5<br>10<br>22<br>85<br>108<br>129<br>100 | 85 | 85 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 7<br>3<br>5<br>7<br>9<br>11<br>13<br>15<br>10 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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V2

| | |
|---|---|
| **Started on** | Friday, 28 August 2026, 9:25 AM |
| **State** | Finished |
| **Completed on** | Friday, 28 August 2026, 9:28 AM |
| **Time taken** | 3 mins 27 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int findPair(int arr[],int left, int right,int x){
    if(left >= right)
        return 0;
    int sum = arr[left]+arr[right];
    if(sum ==x){
        printf("%d\n",arr[left]);
        printf("%d\n",arr[right]);
        return 1;
    }
    if(sum > x)
       return findPair(arr,left,right-1,x);
    return findPair(arr,left+1,right,x);

}
int main(){
    int n,x;
    scanf("%d", &n);
    int arr[n];
    for(int i=0;i<n;i++)
       scanf("%d",&arr[i]);
    scanf("%d",&x);
    if(!findPair(arr,0,n-1,x))
       printf("No\n");
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

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Marks for this submission: 1.00/1.00.

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V2

| | |
|---|---|
| **Started on** | Friday, 28 August 2026, 9:29 AM |
| **State** | Finished |
| **Completed on** | Friday, 28 August 2026, 9:35 AM |
| **Time taken** | 6 mins 38 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>

void quickSort(int arr[], int low, int high)
{
    if (low < high)
    {
        int i = low, j = high;
        int pivot = arr[(low + high) / 2];
        int temp;

        while (i <= j)
        {
            while (arr[i] < pivot)
                i++;
            while (arr[j] > pivot)
                j--;
            if (i <= j)
            {
                temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
                i++;
                j--;
            }
        }

        if (low < j)
            quickSort(arr, low, j);
        if (i < high)
            quickSort(arr, i, high);
    }
}

int main()
{
    int n;
    scanf("%d", &n);
    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    quickSort(arr, 0, n - 1);

    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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